

WARNEMUENDE, Germany

WMO: 101700

Lat: 54.1803N

Lon: 12.0808E

Elev: 42

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.5	19.0	8.5	8.5	15.2	13.2	10.8	20.0	35.3	38.4	32.6	37.3	7.4	110	0.642

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	79.6	64.8	75.7	63.9	72.0	62.8	67.2	75.2	65.5	72.9	64.0	70.5	8.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.3	90.3	70.2	62.6	85.0	68.9	61.2	80.8	67.4	31.6	75.4	30.4	73.2	29.2	70.7	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.3	27.7	24.3	DB	14.5	88.4	7.9	4.3	8.8	91.4	4.2	93.9	-0.2	96.3	-5.9	99.5	(4)
(5)			WB	13.5	71.7	7.4	2.3	8.2	73.4	3.9	74.8	-0.2	76.1	-5.6	77.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.2	34.1	33.7	39.2	44.7	53.5	59.3	63.7	63.1	57.3	50.3	41.4	36.9		
	DBStd	12.24	8.60	7.46	6.23	5.86	6.01	5.38	4.61	4.66	4.14	5.65	5.46	5.98		
	HDD50	2229	493	457	337	180	28	1	0	0	1	64	261	407		
	HDD65	6236	958	876	799	608	362	186	81	95	234	456	709	872		
	CDD50	1568	0	1	3	22	135	280	424	406	221	73	2	1		
	CDD65	100	0	0	0	0	4	16	40	36	4	0	0	0		
	CDH74	538	0	0	0	1	37	99	192	189	20	0	0	0		
	CDH80	119	0	0	0	0	2	24	44	47	2	0	0	0		
(14) Wind	WSAvg	10.6	12.2	10.9	10.5	9.8	9.0	9.8	10.8	9.5	11.0	10.5	11.1	11.8		
(15) Precipitation	PrecAvg	23.7	1.9	1.3	1.5	1.5	1.9	2.5	2.8	2.5	2.1	1.8	1.9	2.0		
	PrecMax	37.6	3.7	3.3	3.9	5.3	4.5	6.1	13.5	7.6	6.3	4.2	5.1	4.1		
	PrecMin	16.5	0.0	0.1	0.2	0.0	0.2	0.0	0.2	0.4	0.5	0.3	0.2	0.4		
	PrecStd	4.6	0.9	0.7	0.8	1.0	1.0	1.3	2.0	1.6	1.2	0.9	1.0	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.4	56.3	60.2	69.6	78.8	83.9	85.4	85.1	76.9	69.0	55.1	52.9		
		MCWB	48.6	50.9	51.4	55.3	63.8	65.9	66.8	66.8	65.2	61.9	51.6	49.6		
	2%	DB	48.0	50.3	53.3	62.1	72.7	76.1	78.7	79.2	70.7	64.0	52.6	50.3		
		MCWB	45.8	46.2	47.4	52.4	59.5	63.8	65.1	64.9	60.7	57.8	50.1	47.9		
	5%	DB	46.3	45.8	49.7	56.6	67.0	71.0	74.9	73.9	66.4	60.4	50.4	47.9		
		MCWB	44.3	42.7	45.8	49.7	56.9	62.0	64.4	64.2	59.4	55.6	48.4	46.1		
	10%	DB	44.0	42.9	47.0	53.0	62.4	67.7	71.4	70.1	63.7	57.6	48.6	44.9		
		MCWB	41.9	40.2	43.7	47.9	55.0	60.4	63.5	62.6	58.3	53.8	46.6	43.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.1	51.5	52.4	57.1	63.9	67.7	70.0	69.7	66.1	63.1	52.5	50.3		
		MCDB	50.8	55.4	59.1	67.2	75.4	77.5	78.4	79.9	75.1	68.3	54.2	52.0		
	2%	WB	46.0	46.7	48.5	53.1	60.8	65.1	67.6	66.6	62.9	59.0	50.3	48.1		
		MCDB	47.3	50.0	52.2	60.3	69.1	73.2	75.1	74.2	68.2	62.5	52.0	50.0		
	5%	WB	44.3	43.2	46.2	50.3	58.0	63.3	65.7	65.1	60.9	56.3	48.6	45.9		
		MCDB	46.2	45.7	49.1	55.9	65.7	70.0	72.8	71.6	64.6	59.5	50.3	47.8		
	10%	WB	42.1	40.6	43.9	48.1	55.4	61.3	64.2	63.5	58.9	54.4	47.0	43.4		
		MCDB	43.8	42.6	46.7	52.8	60.8	67.0	70.3	69.3	62.6	57.1	48.5	44.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	7.1	8.9	10.3	11.9	11.1	11.4	11.7	10.3	9.4	7.1	6.3		
		MCDBR	8.2	10.1	13.3	17.5	20.7	18.4	18.8	19.4	15.9	13.4	8.7	8.7		
	5% WB	MCWBR	7.6	8.3	9.5	10.4	10.5	9.6	9.3	9.3	9.2	8.8	7.9	8.3		
		MCWBR	7.9	10.2	12.3	16.4	18.0	16.5	15.7	15.8	13.9	11.9	8.2	8.3		
(40) Clear-Sky Solar Irradiance	taub		0.319	0.341	0.372	0.400	0.393	0.391	0.410	0.409	0.377	0.362	0.339	0.311		
		taud	2.315	2.318	2.263	2.255	2.332	2.362	2.327	2.342	2.412	2.407	2.365	2.306		
	Ebn at Noon		199	235	252	262	271	272	264	257	250	226	190	175		
		Edh at Noon	18	25	33	38	37	36	37	35	29	23	18	15		
(44) All-Sky Solar Radiation	RadAvg		164	364	763	1305	1657	1763	1639	1340	969	508	210	116		
	RadStd		21	59	73	161	187	127	200	98	94	68	25	10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (55 neighbors)	+0.68	N/A	N/A	N/A	N/A	N/A	N/A	-281	N/A	N/A

Nomenclature: See separate page